# EzVision Campus — System Spec & Smart Campus Audit

Oct 26, 2026 · @Rohenn

## Summary

No Malaysian university, public or private, runs AI on its hostels today, and that is EzVision's opening. Of 38 universities audited, only 4 have a confirmed AI or vision security system: UM and Sunway (library face recognition), APU (plate parking) and UMPSA (Intelligent Operation Centre, May 2026). The rest have cameras watched by guards, RFID stickers and card-access hostels.

- **The world leaders** (ASU, Michigan State, Binghamton, NTU, ITS Surabaya) all run one command centre that joins cameras, plates, access and evidence. Malaysian campuses have the cameras but not the alerts.
- **Several universities already require analytics on paper.** USM, UTHM and UMPSA wrote "video analytics" into their CCTV policies; EzVision can deliver what those policies already ask for.
- **EzVision Campus** is one AI layer over existing cameras: six modules, led by **Hostel Integrity** for opposite-gender entry, overstays and fire-exit use, then Safe Campus, Mobility (ParqOS), Welfare (AIWAS), HSE and Space & Admin.
- **No face recognition by default.** Gender is estimated from appearance only at single-gender block doors, and a warden checks every alert.
- **First targets:** UMPSA, USM, UM and UiTM among public universities; Nottingham Malaysia and Taylor's among private ones. Start with a paid pilot on two hostel blocks, about 20 cameras.

## Audit: Malaysian public universities

Only 2 of 20 public universities have a confirmed live AI or vision security system: UM's library facial recognition and UMPSA's Intelligent Operation Centre. Most sit at "CCTV unit + control room", with guards watching screens. No public university shows AI at hostels, which is where the opposite-gender problem lives.

| University | What's deployed (year) | Maturity | Evidence |
|---|---|---|---|
| UMPSA | Intelligent Operation Centre at Pekan: AI, IoT and analytics for security, traffic, energy, with phased smart CCTV (May 2026). Guards added to hostels after a Jan 2025 incident; CCTV "under consideration" | High | IOC news, hostel incident |
| UM | Facial recognition at library entrance (live); auxiliary police; "Integrated Surveillance" in Smart Sustainable Campus 2022–2030 roadmap | Medium | library, roadmap |
| USM | CCTV policy creates a central control room (SMRC) with "video analytics" (2017, reviewed 2019); no proof analytics run | Medium | CCTV policy |
| UPM | Plate-recognition barrier ("Palang Automatik LPRS") likely live; Huawei campus-network MoU (2015) | Medium (unverified) | APSeC (page blocked) |
| UTM | RFID vehicle stickers, e-Pelekat 2.0 (2025); Perodua smart-mobility pilot (Jan 2026) | Low–Medium | sticker procedure |
| UniSZA | Aimetis video management with motion and perimeter analytics on 40+ cameras, 3 campuses (2011) | Low–Medium | asmag |
| UiTM | Huawei smart-campus MoU (networks, 2020); CCTV used after the fact to check a viral Puncak Alam hostel-intrusion claim (Apr 2024) | Low | SAYS |

| University | What's deployed (year) | Maturity | Evidence |
|---|---|---|---|
| UKM | 40 CCTV cameras per building at one faculty (FKAB); Beam e-scooters | Low | FKAB |
| UTHM | Draft CCTV policy: central control room (CMCC), PTZ video analytics, traffic flow | Low | draft policy |
| UMK | Face, plate and thermal "Smart Security & Surveillance" planned 2020–22; no proof delivered | Low | smart campus plan |
| UMS | Central CCTV unit; Huawei campus network (2017) | Low | CCTV unit |
| UPSI | CCTV unit enforces traffic and college rules (2024) | Low | CCTV function |
| IIUM | Control room; lift-CCTV training (Jul 2026); gate plate recognition is a 2025 research prototype only | Low | OSeM, paper |
| UniMAP | RFID gateless vehicle entry, research system under test (2021) | Low | paper |
| UNIMAS | e-Sticker; 2026 "Smart, Green & Inclusive Campus" priority with no camera plan | Low | VC address 2026 |
| UPNM | CCTV control unit; online vehicle stickers | Low | Security Division |
| UUM, USIM, UMT, UTeM | No public evidence found beyond network deals or sticker systems | Unknown | — |

What the pattern means:

- **Policies promise analytics that do not exist.** USM, UTHM and UMPSA wrote "video analytics" into CCTV policy years ago. EzVision can deliver what the policy already requires.
- **Vehicles use RFID stickers, not plate recognition.** UTM, UPNM, UNIMAS and UniMAP; ParqOS is a direct upgrade.
- **Big-vendor deals are networks, not security.** Huawei (UPM, UMS, UiTM) and TM ONE (UUM) sold Wi-Fi and data centres. No Hikvision, Dahua or Tapway campus security deal is public.
- **Incidents bring guards first, cameras later.** UMPSA (2025) and UiTM Puncak Alam (2024) show the buying trigger is a hostel incident.

## Audit: Malaysian private universities

Only 2 of 18 private universities have verifiable AI or vision security: Sunway (library facial recognition, 2019) and APU (plate-recognition parking). Taylor's claims "AI-CCTV" in its residences, but with no detail. Everywhere else, hostels run on access cards, guard-house visitor books and plain CCTV reviewed after incidents.

| University | What's deployed (year) | Maturity | Evidence |
|---|---|---|---|
| APU | APSPark plate-recognition parking with pre-registered vehicles and cashless APCard; residence access cards, 24/7 security | High (parking) | parking, residence |
| Sunway | Facial recognition at library entry and exit, replacing cards (2019, still in use) | Medium–High | SAYS, Business Today 2023 |
| Taylor's | Residence FAQ claims "AI-CCTV surveillance covering all floors and entrances"; turnstiles, 24h guards | Medium (claim) | residence FAQ |

| University | What's deployed (year) | Maturity | Evidence |
|---|---|---|---|
| Nottingham Malaysia | 300+ CCTV cameras; ID-card residence doors; app-based parking permits (2023) | Medium | accommodation handbook |
| Xiamen Univ. Malaysia | Campus e-card for doors, library, payments; privacy notice already covers CCTV and biometric data; visitors register by hand | Medium | e-card, PDP notice |
| MSU | ID-card parking gates; Touch 'n Go visitor parking | Low–Medium | parking |
| Heriot-Watt Malaysia | Cashless parking run by KLCC Parking Management | Low–Medium | transport |
| Curtin Malaysia | Residence cards; CCTV reviewed after incidents; guard-house visitor log; stickers and clamping (2025) | Low | residence rules |
| UTAR | Guarded gates, numbered stickers, visitor passes, clamping | Low | vehicle stickers |
| MMU, UNITEN, INTI, MAHSA, Swinburne Sarawak, UCSI | Access cards, guards and plain CCTV only | Low | MMU, INTI |
| Monash Malaysia, UTP, SEGi | No public evidence found | Unknown | — |

Most likely private buyers: **Nottingham** (300+ cameras, no analytics), **Taylor's** (already sells safety as "AI-CCTV"), **Sunway** (early adopter; Sunway City already runs plate parking), **APU** (plate and card base to build on) and **XMUM** (privacy notice already covers biometrics).

# Audit: global smart-campus leaders

The leaders all bought one command centre joining cameras, access, plates and evidence, then staffed it close to 24/7. ITS Surabaya is the closest regional model: face plus plate gates for motorcycles, speed cameras, and a command centre.

| University | What's deployed | Vendor | Year | Result or note |
|---|---|---|---|---|
| ITS Surabaya (Indonesia) | 76 face-access terminals; 10 radar plate cameras that send speeding notices over 40 km/h; 42 barriers open only when rider face and plate match; command centre with map video wall | Hikvision | 2025 | Vendor reports motorcycle theft and speeding fell (no figures) |
| NTU (Singapore) | New CCTV with video analytics, 24/7 Security Command Centre; halls and staff housing first | Certis | 2024 | Goals only, no results published |
| NTU, NUS, SMU, SUTD (Singapore) | Anti-voyeurism: motion alarms in hall toilets, CCTV at restroom entrances only, controlled toilet access | Not named | 2021 | Layered measures; no cameras inside |
| Arizona State (US) | Real Time Analysis Center: 3,000–4,000+ cameras, plate readers, body cams, live 911 transcription | Genetec | 2023 | Property and bike theft "decreased significantly"; national awards |

| University | What's deployed | Vendor | Year | Result or note |
|---|---|---|---|---|
| Binghamton (US) | 1,635 cameras, 4,800+ readers, plate readers, crowd-gathering analytics, evidence management | Genetec | 2025 | Faster incident response (not quantified) |
| Michigan State, Grand Canyon (US) | MSU ~3,000 cameras in a 24/7 operations centre with clothing and direction search; GCU 500+ AI cameras linked to dispatch | Genetec, Verkada | 2025 | "Force multiplier"; no metrics |
| Chung-Ang (South Korea) | Cloud cameras; search by clothing, vehicle or plate; cross-camera timeline; face blur | Verkada | n/a | Tracked one suspect across campus with police |
| Curtin (Australia) | AI that learns "normal" scenes and flags anomalies to operators | icetana | 10+ yrs; renewed 2026 | Licences more than doubled; contract renewed to 2029 |
| Boston College (US) | 16 AI walk-through weapons screeners at stadium and arena | Evolv | 2023 | Events only |
| Tsinghua, XJTU, Zhejiang (China) | Face scan at enrolment, labs, smart classrooms | Uniview | ~2019 | Followed by the Ministry of Education's 2019 pledge to curb face recognition in schools (The China Project) |

| University | What's deployed | Vendor | Year | Result or note |
|---|---|---|---|---|
| KAUST (Saudi Arabia) | 15.4 km perimeter with automatic intrusion detection; CCTV and drones with a published privacy notice | SightLogix, Geutebruck | 2009 | Old project |

Two governance lessons:

- **UCLA dropped face recognition in 2020** after a test falsely matched 58 athletes and staff to mugshots (Security Sales). Classroom "attentiveness" tracking in China drew a ministry crackdown.
- **Edinburgh publishes the template:** a privacy impact assessment for every camera, privacy masking, 30-day retention, named viewer roles, logged access and quarterly audits (CCTV Protocol).

## Gap analysis and target list

Malaysian campuses are roughly where ASU and Binghamton were before their command centres: cameras exist, but nobody gets an alert. Hostel AI is the biggest gap: no Malaysian university, public or private, shows any.

| Capability | Global leaders | Malaysian public | Malaysian private | EzVision today |
|---|---|---|---|---|
| Command centre with live alerts | Standard (ASU, MSU, NTU, ITS) | UMPSA only; others watch screens | None | Console exists (ParqOS, Condo) |
| Plate recognition at gates | Standard | UPM likely; others RFID stickers | APU only | Live |
| Hostel entry analytics | NTU halls first; ITS face gates | None | Taylor's claims only | Build (this spec) |
| Fight, bullying, vape alerts | Verkada fight; nobody does vape | None | None | PoC (AIWAS) |

| Capability | Global leaders | Malaysian public | Malaysian private | EzVision today |
|---|---|---|---|---|
| Search by clothing or plate | Standard | None | None | Build |
| Face recognition | Common in China and ITS; dropped at UCLA | UM library | Sunway library | Opt-in only |
| Published privacy rules | Edinburgh template | CCTV policies at USM, UTHM, UMPSA | XMUM privacy notice | Ship as standard |

Target list, in order:

| # | University | Why now | Entry module | Who to approach |
|---|---|---|---|---|
| 1 | UMPSA | Built an Intelligent Operation Centre in May 2026 and had a hostel incident in 2025; needs analytics to feed it | Hostel Integrity + Safe Campus | IOC team, Bahagian Keselamatan |
| 2 | USM | Its own CCTV policy requires video analytics in the SMRC control room | Safe Campus | Bahagian Keselamatan |
| 3 | UM | Smart Sustainable Campus roadmap lists Integrated Surveillance; alumni link | Hostel Integrity | Smart campus office, HEP |
| 4 | UiTM | Largest student population and hostel estate; Puncak Alam intrusion scare (2024) | Hostel Integrity | HEP, Bahagian Keselamatan |
| 5 | UTHM | Draft CCTV policy with a planned control room (CMCC) | Safe Campus | Bahagian Keselamatan |

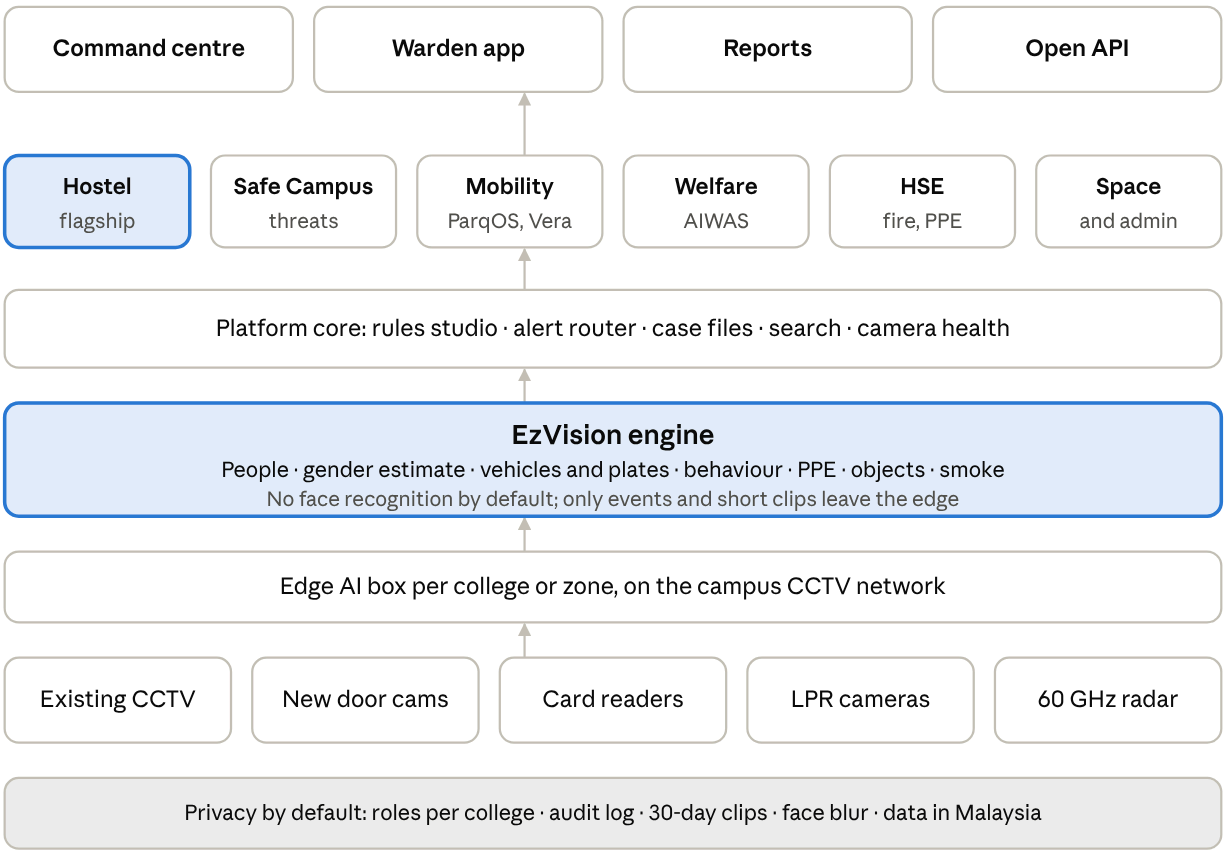
| # | University | Why now | Entry module | Who to approach |
|---|---|---|---|---|
| 6 | Nottingham Malaysia | 300+ cameras with no analytics | Safe Campus | Campus security |
| 7 | Taylor's | Already markets "AI-CCTV"; needs it to be real | Hostel Integrity | Residence office |
| 8 | UTM, UPNM, UNIMAS | RFID stickers ready for upgrade to plate recognition | Mobility (ParqOS) | Bahagian Keselamatan |

Caveat: the search covered English and Malay web pages, but Malaysian news sites search poorly. Confirm the "Unknown" universities by calling their security offices.

## System overview

EzVision Campus is one AI layer over the university's existing cameras: edge boxes analyse video on campus, one engine detects, the platform core applies rules, and six modules route alerts to the right office.

**Six campus modules share one engine, one core and one console**




EzVision Campus architecture · 7 layers

Data flows up from cameras and sensors to the people who act. Hostel Integrity is highlighted because it leads the sale; the other five modules reuse the same engine, core and console.

## Modules and features

EzVision Campus has six modules on one platform core. Hostel Integrity leads the sale; the others reuse the same engine and console.

| Module | Features | Main users | Status |
|---|---|---|---|
| **1. Hostel Integrity** (flagship) | Block and zone mapping; opposite-gender boundary alerts; visitor beyond the lounge; overstay past visiting hours; after-midnight entry; side-door and fire-exit entry; tailgating vs. card taps; delivery riders past drop-off; weekly breach report by block, door and hour | Wardens, felo, Pengetua, HEP | Build |
| **2. Safe Campus** | Fight; person down; crowd build-up or sudden run; loitering; after-hours intrusion; person at height or water edge; person or plate of interest; search by clothing, colour or plate; cross-camera tracking; weapon and unattended bag (new models) | Command centre, auxiliary police | PoC + Build |
| **3. Mobility** (ParqOS, Vera) | Plate-recognition gates replacing RFID stickers; visitor plate pre-registration; parking occupancy; illegal and OKU-bay parking; helmet compliance; speeding; blacklisted vehicles; barrier-free visitor payment | Traffic unit, parking office | Live + Build |
| **4. Welfare** (AIWAS) | Bullying and ragging; vaping and smoking; vandalism; unsafe night routes; wellbeing signals routed to counselling, never to discipline | HEP, counselling | PoC |
| **5. HSE** | Smoke and fire; lab and construction PPE; flash flood; radar falls in toilets and changing rooms; animal intrusion (monkeys, dogs) | Facilities, OSH | Live + New |

| Module | Features | Main users | Status |
|---|---|---|---|
| **6. Space & Admin** | Room and library occupancy; cafeteria queues; event crowd counts; empty-room energy saving; lost-item search; open-day analytics | Facilities, academic affairs | Build |

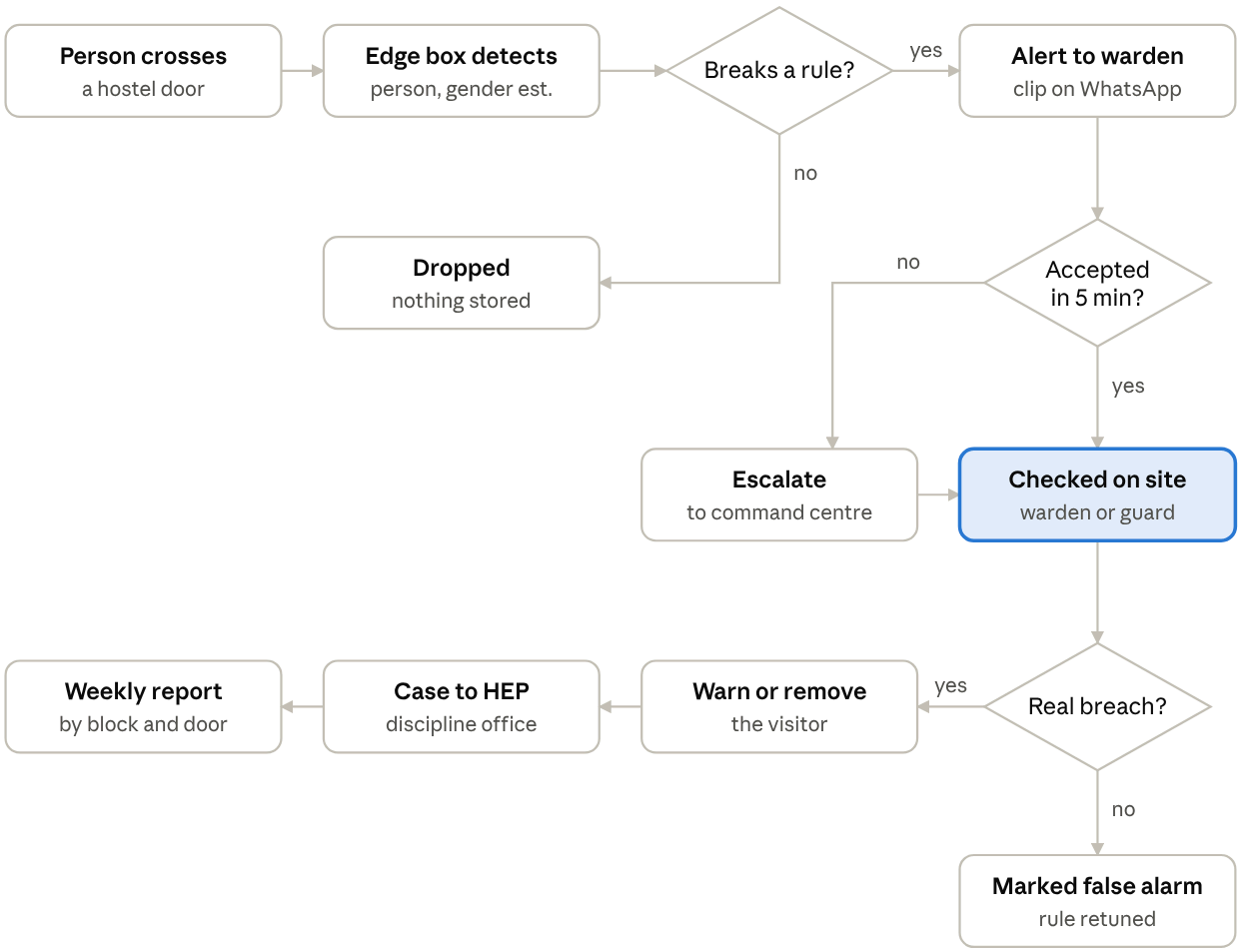
The platform core, shared by every module:

- **Camera onboarding:** any brand over RTSP/ONVIF; draw zones and lines on each view.
- **Edge inference:** video is analysed on campus; only events, snapshots and short clips leave the box.
- **Rules studio:** zone + object + condition + schedule + action, set by the university without code.
- **Alert router:** routes by duty roster, escalates when nobody acknowledges.
- **Case management:** link alerts, add notes, record the outcome, export evidence with an audit trail.
- **Camera health:** offline, blocked, blurred or moved cameras flagged daily.
- **Reports:** weekly and monthly PDF summaries for the Deputy Vice-Chancellor (TNC HEP) and security director.

## Hostel Integrity: how an alert flows

The flagship module catches opposite-gender entry, visitors past the lounge, overstays, after-midnight entries and fire-exit use. A person checks every alert before anything happens to a student.

**Every hostel alert ends with a person deciding**




Hostel Integrity alert flow · 3 decisions

Detections that break no rule are dropped on the edge box. An unaccepted alert escalates to the command centre after 5 minutes, and false alarms retune that camera's rule.

Rules per block, set by the Pengetua in the rules studio:

| Rule | Active |
|---|---|
| Other gender past the lounge line | Always, in single-gender blocks |
| Visitor past the lounge line | Always |
| Visitor still inside at closing | After visiting hours end (e.g. 10 pm) |
| Any entry | After curfew (e.g. 12 midnight) |

| Rule | Active |
|---|---|
| Entry through exit-only door or fire exit | Always |
| More people than card taps | Always |

Hours follow the UTeM rule book as an example: visiting 5–10 pm on weekdays, midnight curfew (UTeM). Each university sets its own.

## Users, roles and screens

Ten roles, each seeing only its own module and area. A warden sees one college; the command centre sees the whole campus.

| Role | Office | Sees | Can do |
|---|---|---|---|
| Command centre operator | Bahagian Keselamatan | All live alerts, all cameras | Acknowledge, dispatch guards, open cases |
| Security director | Bahagian Keselamatan | Everything, plus reports | Approve evidence exports, change rules |
| Warden or felo on duty | Kolej Kediaman | Hostel alerts for their college only | Accept, verify on site, close with outcome |
| Pengetua (college principal) | Kolej Kediaman | Their college's breach reports | Set visiting hours and curfew per block |
| HEP discipline officer | Student affairs | Confirmed cases only, not live video | Open disciplinary action |
| Counsellor | Counselling unit | Wellbeing alerts only | Follow up with the student |
| Facilities and OSH officer | Facilities | HSE and Space alerts and dashboards | Dispatch maintenance |
| Traffic and parking officer | Traffic unit | Mobility alerts, plate lists | Manage permits, issue notices |
| University DPO | Legal or registrar | Audit log, exports, retention settings | Review access, handle data requests |

| Role | Office | Sees | Can do |
|---|---|---|---|
| System admin | IT and EzGIG | Configuration, camera health | Add cameras, users and roles |

Screens:

1. **Command centre console:** campus map, live alert queue sorted by severity, video wall.
2. **Alert detail:** 10-second clip, snapshot, rule that fired, camera on the map, actions.
3. **Warden app (mobile):** alert inbox, on-duty toggle, accept and verify, outcome in two taps, block breach history.
4. **Search and investigation:** search by time, camera, clothing colour, object or plate; cross-camera timeline.
5. **Case file:** linked alerts, notes, outcome, evidence export with watermark and audit entry.
6. **Hostel dashboard:** breaches by block, door and hour; trend over the semester.
7. **Rules studio:** draw zones and lines, set schedules and thresholds, preview against yesterday's footage.
8. **Camera health:** offline, blocked, blurred or moved cameras.
9. **Reports:** weekly and monthly PDFs for management.
10. **Admin and privacy:** users, roles, retention, audit log, data requests.

## Data model, rules engine and alert lifecycle

The system stores events and alerts, not identities. Raw detections live seconds on the edge box; only alerts with a short clip are kept.

| Entity | Key fields | Kept for |
|---|---|---|
| University, Campus | Name, address, time zone, retention policy | Contract term |
| Block | Campus, type (hostel female, hostel male, academic, office, car park), visiting hours, curfew | Contract term |
| Zone | Camera, polygon or line, type (entrance, lounge, corridor, exit-only, restricted), block | Contract term |

| Entity | Key fields | Kept for |
|---|---|---|
| Camera | Stream URL, edge box, zones, position on map, health | Contract term |
| Edge box | Site, cameras, model versions, status | Contract term |
| Rule | Zone, object, condition, schedule, severity, route, escalation | Versioned forever |
| Detection | Time, camera, object class, attributes, track ID, confidence | Seconds (edge only) |
| Alert | Rule, detections, snapshot, clip, status, assignee, timestamps | 30 days default |
| Case | Linked alerts, notes, outcome, exports | Per university policy |
| Vehicle permit, visitor pass | Plate, holder type, validity, host | Until expiry + 30 days |
| User, role, duty roster | Name, office, scope, shift | While active |
| Audit log | Who viewed, exported or changed what, when | 7 years |

A rule is written in the rules studio like this:

```
WHEN   person (gender_est = male, confidence ≥ 0.8)
CROSSES line "Blok C main door" INTO zone "residential"
DURING 00:00–24:00 daily      // female block, no male entry
THEN   alert severity HIGH → warden on duty, Kolej Tun Fatimah
ESCALATE to command centre if not acknowledged in 5 min
```

Every alert moves through the same steps:

1. **New:** the rule fires; clip and snapshot attached.
2. **Acknowledged:** a warden or operator accepts it; the escalation timer stops.
3. **Verified:** checked on site or on video; marked true, false alarm or unsure.
4. **Actioned:** for true alerts, the action taken (warning, removed visitor, HEP referral).
5. **Closed:** outcome recorded. False alarms feed back to tuning, so thresholds improve per camera.

The college name in the example is illustrative.

## Hardware and deployment

Reuse existing cameras wherever they cover a doorway at head-and-shoulder angle; add cameras only at side doors and fire exits, which most hostels lack. All figures below are planning estimates to be confirmed in a site survey.

Per hostel block:

| Position | Cameras | Notes |
|---|---|---|
| Main entrance, inside facing in and out | 2 | 2.5–3 m height, head-and-shoulder view, infrared for night |
| Lounge (ruang tamu) to corridor boundary | 1 | The line visitors must not cross |
| Each side door, stairwell door and fire exit | 1 each | Usually 2–4 per block; often new cameras |
| Corridors, one per floor (optional) | 0–4 | Only where overstay tracking needs it |
| **Typical total** | **6–10** | |

Site components:

- **Edge AI box:** GPU appliance in the block's or college's network room. Stream capacity per box depends on model and resolution; size it in the pilot.
- **Network:** existing CCTV VLAN; the AI reads the camera's sub-stream (720p) so bandwidth stays low.
- **Storage:** alert clips on the edge box for 30 days; continuous recording stays on the university's existing NVR.
- **Command centre:** a web console; no special hardware beyond screens.
- **Hosting:** on-premises in the university data centre, or hybrid with metadata in a Malaysian cloud region.

Illustrative sizing:

| Stage | Scope | Cameras analysed |
|---|---|---|
| Pilot | 2 hostel blocks (one female, one male) + 1 main gate | About 20 |
| Phase 1 | All hostel blocks at one campus (assume 20 blocks) | About 160–200 |
| Full campus | Hostels + gates + car parks + academic buildings | 600–1,000+ |

## Integrations

EzVision Campus plugs into what the university already runs; only the camera feed is required, the rest add accuracy.

| System | Why | How | Needed for pilot? |
|---|---|---|---|
| Existing cameras and NVR (Hikvision, Dahua, others) | Video source | RTSP / ONVIF sub-stream | Yes |
| Existing VMS (HikCentral, Milestone, Genetec) | Show EzVision alerts inside the console operators already use | Event API or plug-in | No |
| Hostel card readers | Count card taps vs. people for tailgating | Door-event export or controller API | No |
| Student information system | Which block each resident belongs to | Nightly file or API, block assignment only | No |
| Visitor registration | Known visitors and plates; flags walk-ins | API, or EzVision's own visitor pass | No |
| e-Pelekat / RFID sticker database | Permit list for plate recognition | Export or API | Mobility only |
| ParqOS | Gates, parking occupancy, payment | Native | Mobility only |

| System | Why | How | Needed for pilot? |
|---|---|---|---|
| Building management (BMS) | Turn off AC and lights in empty rooms | BACnet | Space only |
| Fire alarm panel | Confirm smoke alerts with the panel | Dry contact or API | HSE only |
| WhatsApp Business, Telegram, SMS | Alerts to wardens and guards | Messaging API | Yes |
| University sign-in (Microsoft Entra, Google) | Single sign-on, leavers removed automatically | SAML / OIDC | No |
| HEP discipline system | Hand confirmed cases to student affairs | Case export (PDF) or API | No |
| PDRM requests | Evidence handover for police cases | Watermarked export with audit entry | No |

## Security, privacy and PDPA

Privacy is built into the defaults so the university's DPO can approve the system quickly. Under the amended PDPA, biometric data is sensitive personal data; DPO and breach-notification duties apply from 1 June 2025; fines reach RM1,000,000; and processors such as EzGIG carry security duties directly (Data Protection Report).

| Control | Default | Why |
|---|---|---|
| Face recognition | Off; opt-in for staff in restricted labs only | Biometric data is sensitive; UCLA dropped it over false matches |
| Gender estimation | Only at single-gender block boundaries, during restricted hours | Purpose is to enforce published hostel rules; names no one |
| Classroom attention scoring | Not offered | Unreliable and the clearest red line globally |

| Control | Default | Why |
|---|---|---|
| Toilets and changing rooms | No cameras; radar only | Singapore universities keep cameras at entrances only |
| Retention | 30 days for alert clips; cases per university policy | Edinburgh's protocol uses 30 days |
| Privacy masking | Faces blurred in exports unless the security director unblurs for a case | Minimises data shared |
| Access | Role-based, scoped by college or office; single sign-on | A warden sees one college only |
| Audit | Every view, export and rule change logged for 7 years | DPO review and police chain of custody |
| Data residency | Edge on campus; metadata in Malaysia | PDPA transfer rules |
| Encryption | In transit and at rest | Processor security duty |
| Human decision | AI never issues a penalty; a person verifies every alert | Fairness and false-alarm protection |

EzGIG ships with every deployment: a privacy impact assessment template, signage text in Malay and English, a breach-response runbook, an annual penetration test, and a bias test of the gender and behaviour models across hijab, hoodies, long hair and night infrared.

## Rollout, pilot KPIs and open questions

Start with one paid pilot on two hostel blocks, prove the numbers, then expand block by block. Each phase passes a gate before the next starts.

1. **Survey (2 weeks).** Walk both blocks, list cameras and blind doors, agree rules with the Pengetua and DPO. *Gate:* signed rules sheet and privacy assessment.
2. **Pilot build (4 weeks).** Install the edge box and missing door cameras, map zones, connect WhatsApp alerts. *Gate:* all cameras online, test alerts reach the warden.
3. **Shadow run (2 weeks).** Alerts go to EzGIG only, to tune thresholds at night. *Gate:* false alarms under the agreed limit.

4. **Live pilot (8 weeks).** Wardens receive alerts and verify them. *Gate:* KPIs below met; management sign-off.
5. **Phase 1 (one semester).** All hostel blocks at the campus, plus Safe Campus rules in the command centre.
6. **Phase 2 onward.** Mobility at gates, Welfare, HSE, Space & Admin, then other campuses.

Pilot KPIs to agree with the university:

| KPI | How measured | Target to set with client |
|---|---|---|
| Alert precision | % of alerts the warden marks true | Agree in survey |
| False alarms per block per night | Alerts marked false | Agree in survey |
| Time to acknowledge | Alert to warden accept | Under 5 minutes |
| Breach trend | Confirmed breaches per week, weeks 1–8 | Falling |
| Camera uptime | % of cameras online daily | 98% or better |
| Warden adoption | % of alerts closed with an outcome | 90% or better |

Open questions:

- [ ] **Gender estimation accuracy** at night and with hijab: test on pilot footage before promising a number.
- [ ] **AIWAS IP:** confirm EzGIG can reuse the bullying and vape models outside Micropay's school deal.
- [ ] **Pilot university:** UMPSA, USM or UM first (see target list).
- [ ] **Commercial model:** outright, rental or per-camera subscription; reuse the ParqOS deal menu.
- [ ] **Hardware:** who supplies the extra door cameras, EzGIG or the university's CCTV contractor.
- [ ] **Consent and signage:** how each university informs residents before go-live.

## Sources

Every university row above links to the page it came from. Additional sources:

- Data Protection Report — Malaysia PDPA Amendment Act 2024
- UTeM — Buku Peraturan dan Tatatertib Kolej Kediaman
- Security Sales — UCLA abandons facial recognition plan
- University of Edinburgh — CCTV Protocol
- EdTech Magazine — Real-time security centers on campus (May 2026)
- EzVision for Universities — Use Case Map — the 61 use cases this spec builds on

Audit method: public web pages in English and Malay, opened and checked on 26 Oct 2026. Some university pages blocked automated reading (UPM, UTeM, UniSZA, UMS), and Malaysian news sites search poorly; "Unknown" means no public evidence, not proof of absence.